ASSIGNMENT NO. : 03

PRACTICAL: Forking an Open-Source AI Project and Submitting a Pull Request

Step 1: Select an Open-Source AI Project

1. Log in to GitHub.
2. Search for an open-source AI project.
3. Open the project repository.
4. Click the **Fork** button.

Output:

```
Forked repository successfully created in your GitHub account.
```

Step 2: Clone the Forked Repository

Copy the URL of the forked repository and clone it.

```
git clone https://github.com/<username>/ai-project.git
cd ai-project
```

Output:

```
Cloning into 'ai-project'...
done.
```

Step 3: Check Existing Branches `git`

```
branch
```

Output:

```
* main
```

Step 4: Create a Feature Branch

Create a separate branch for making the required changes.

```
git checkout -b feature-improvement
```

Output:

```
Switched to a new branch 'feature-improvement'
```

Step 5: Make Changes to the Project

Modify or add the required AI project file.

For example, create a documentation file: `echo`

```
"AI Feature Improvement" > feature.md
```

Check the changes:

```
git status
```

Output:

```
Untracked files:
feature.md
```

Step 6: Add and Commit the Changes `git`

```
add .

git commit -m "Added AI feature improvement"
```

Output:

```
[feature-improvement abc1234] Added AI feature improvement
1 file changed create mode
100644 feature.md
```

Step 7: Push the Feature Branch to GitHub `git`

```
push -u origin feature-improvement
```

Output:

```
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (5/5), done.
  Branch 'feature-improvement' set up to track 'origin/feature-improvement'.
```

Step 8: Create a Pull Request

1. Open the forked repository on GitHub.
2. Select the `feature-improvement` branch.
3. Click **Compare & pull request**.
4. Enter a suitable title and description.
5. Select the original open-source repository as the **base repository**.
6. Select `main` as the base branch.
7. Click **Create pull request**.

Output:

```
Pull request successfully created.
```

Step 9: Review the Pull Request

The project maintainer reviews the submitted changes.

Possible review comments may include:

```
Please update the documentation and
modify the feature according to the
project requirements.
```

Make the requested changes in the local feature branch.

Step 10: Address Review Comments

Modify the required files and commit the changes.

```
git add . git commit -m "Addressed review
```

```
comments"
```

Push the changes:

```
git push
```

Output:

```
Changes successfully pushed to the pull request.
```

Step 11: Resolve Merge Conflicts

If GitHub reports a conflict, update the local main branch:

```
git checkout main
git pull origin main
```

Switch back to the feature branch: `git`

```
checkout feature-improvement
```

Merge the latest main branch:

```
git merge main
```

Output:

```
CONFLICT (content): Merge conflict in feature.md Automatic merge
failed; fix conflicts and then commit the result.
```

Step 12: Resolve the Conflict

Open the conflicted file.

```
<<<<<<< HEAD
Feature branch changes
=======
Main branch changes
>>>>>>> main
```

Manually edit the file and keep the required content. Remove the conflict markers. Then add the resolved file:

```
git add feature.md
```

Step 13: Commit the Conflict Resolution `git`

```
commit -m "Resolved merge conflict"
```

Output:

```
[feature-improvement def5678] Resolved merge conflict
```

Step 14: Push the Conflict Resolution `git`

```
push
```

Output:

```
Changes pushed successfully.
Pull request updated.
```

Step 15: Verify the Pull Request Open the

Pull Request on GitHub and verify that:

- Review comments are addressed.
- Merge conflicts are resolved.
- All required changes are present.
- The pull request is ready for review.

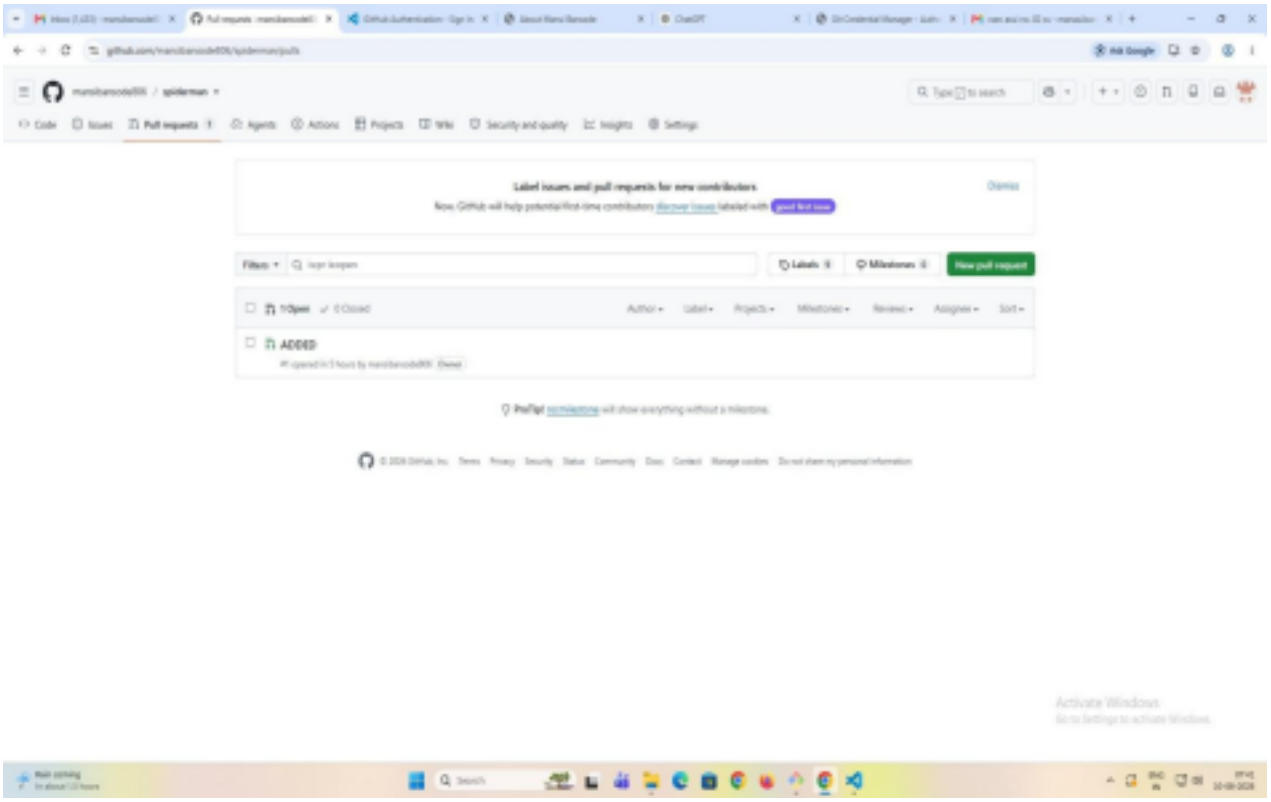
Output:

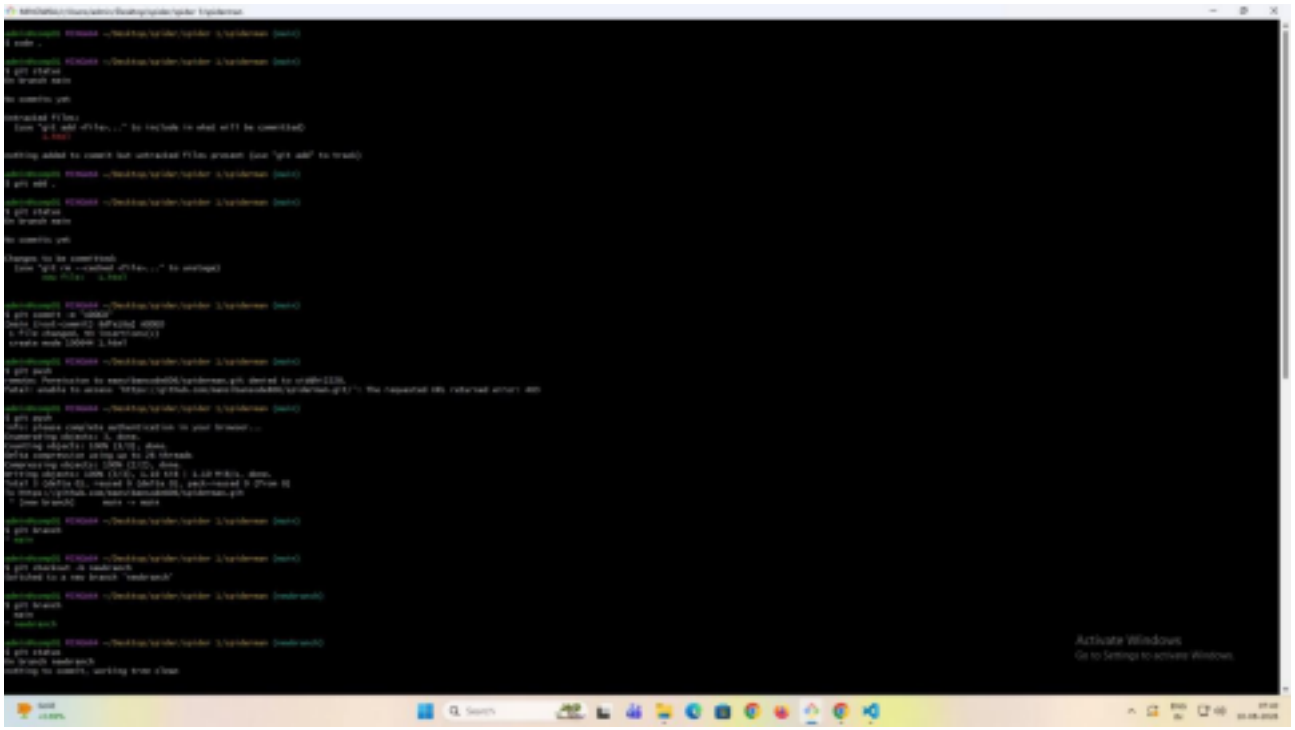
```
Pull Request: feature
improvement → main

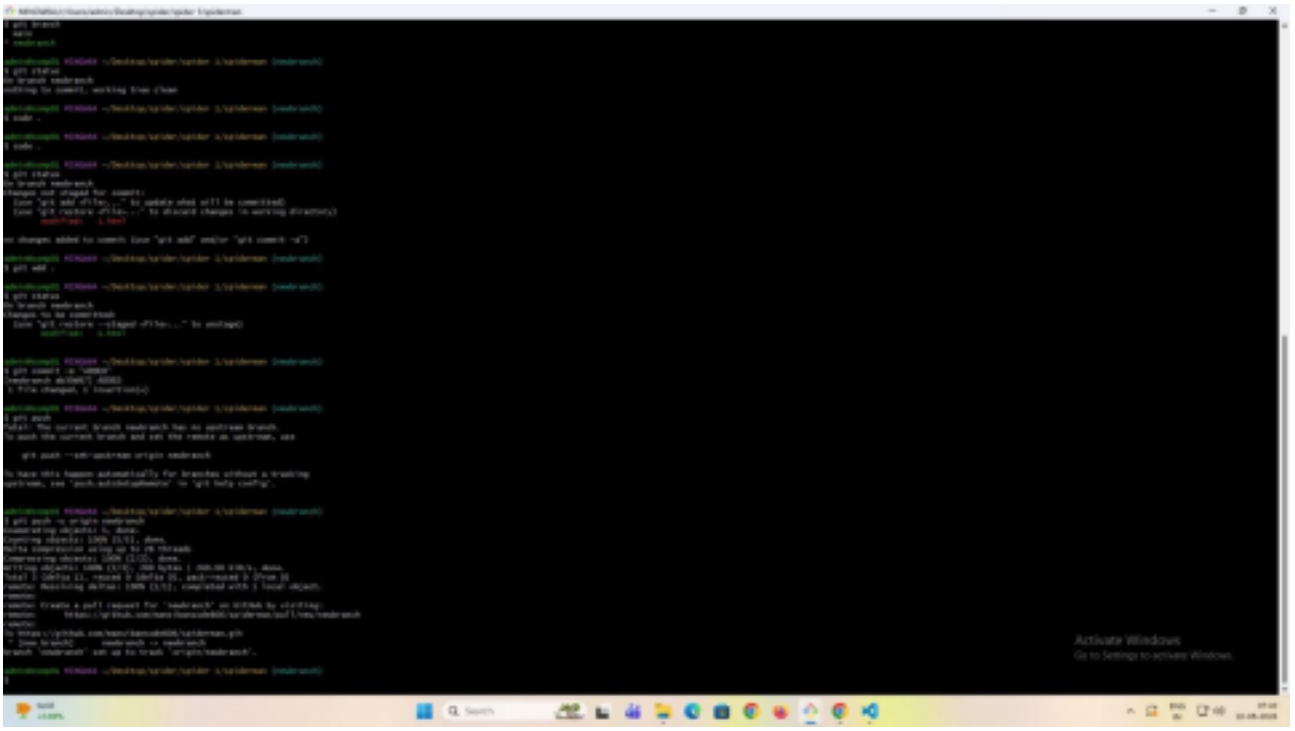
Status: Ready for review
Conflicts: Resolved
```

Result

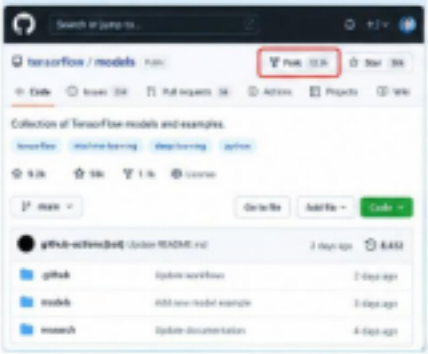
The open-source AI project was successfully forked, cloned, and modified using a separate feature branch. A pull request was created, review comments were addressed, and merge conflicts were successfully resolved.



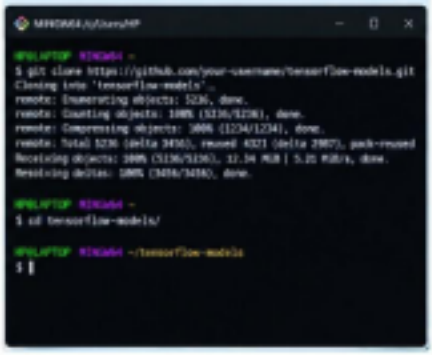




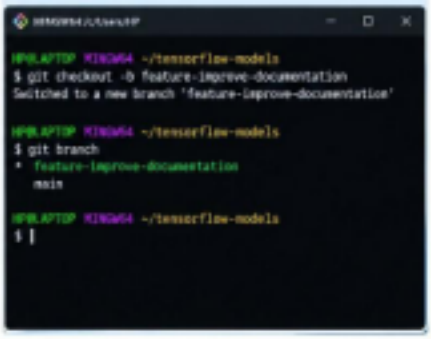
1. GitHub repository showing the Fork button



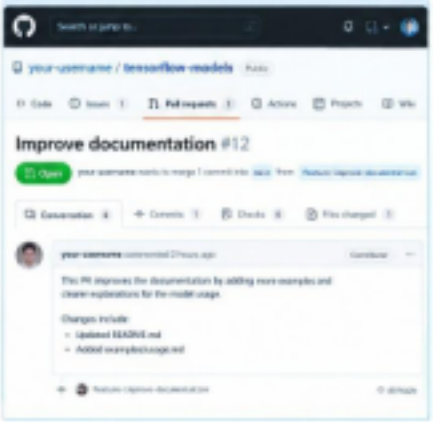
2. Terminal showing git clone and successful cloning



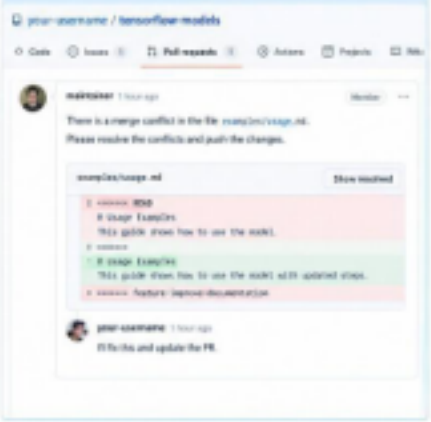
3. Terminal showing creation of feature-improve-documentation



4. GitHub Pull Request page showing the submitted Pull Request



5. GitHub/terminal showing the merge conflict



6. GitHub Pull Request showing the resolved/merged status

